

Integral face–edge duality for knot triangulations

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Abstract

For an ordered ideal triangulation of a knot exterior in S^3 , we give a cochain construction relating its face matrix C to its preferred-longitude gluing matrices A, B . The obstruction to uniqueness is the first cohomology of the ideal compactification. Its vanishing yields nullity doubling and the FAMED matrix identity, including a constrained formulation for singular face matrices. A separate rank argument under a strict angle structure supplies a proposed proof of Wong’s generalized FAMED conjecture. Over the integers, the construction gives a direct-sum decomposition of $\ker_{\mathbb{Z}} B$ and an exact sequence from the two face cokernels to the gluing cokernel, with quotient $\mathbb{Z}/2\mathbb{Z}$. In the unhalved longitude convention this implies $|\det B| = 2|\det C|^2$. We also obtain cyclic face torsion, order-reversal invariance of Smith normal form, and affine face coordinates for fixed lifted gluing systems. The determinant-one assertion and the full Andersen–Kashaev volume conjecture remain open here.

1 Introduction

The gluing equations of an ideal triangulation describe how the shapes of its tetrahedra fit around edges and peripheral curves. In the Teichmüller TQFT construction, a second collection of matrices records relations among variables attached to faces. The FAMED conditions compare these two descriptions. Ben Aribi and Wong introduced these conditions in their study of the Andersen–Kashaev volume conjecture [2]; Ben Aribi, Guilloux and Wong subsequently investigated them computationally [1]. Wong’s generalized conditions include triangulations whose face matrix is singular [3].

The question considered here is why the face matrices and the gluing matrices should satisfy the same constraints. We associate an additive edge cochain to a pair of face assignments, one for each direction of the vertex ordering. Differences of this cochain give the homogeneous shape variables. Face identifications force the edge equations. A separate cocycle on the dual spine shows that the preferred-longitude equation holds as well. The obstruction to recovering the face assignments uniquely is a cohomology group of the ideal compactification; this group vanishes for a knot exterior in S^3 .

This construction gives the rational correspondence below, including singular face matrices. Integral lifts then relate kernels and cokernels over $\mathbb{Z}$. Finally, a combinatorial rank argument under a strict angle structure supplies the additional condition in Wong’s generalized FAMED conjecture. The analytic consequences are applications of Wong’s existing theorem. The determinant-one assertion in the original FAMED conjecture requires a further argument and remains open here.

1.1 Conventions and notation

Let M be the compact exterior of a knot in S^3 , and let $\mathcal{T}$ be an ordered ideal triangulation of its interior with N tetrahedra. All faces are paired; there is one ideal vertex, $2N$ quotient faces and N quotient edges. “Ordered” means that local vertex orders are compatible on glued faces. Fix an orientation of M . The sign of tetrahedron i relative to its vertex order $0 < 1 < 2 < 3$ is $\varepsilon_i \in \{1, -1\}$. If faces opposite vertices k, l in tetrahedra i, j are paired, then

$$\varepsilon_i(-1)^k = -\varepsilon_j(-1)^l. \quad (1)$$

Let X_k be the $N \times 2N$ incidence matrix recording the quotient face opposite local vertex k . Set

$$C = \begin{pmatrix} X_0 - X_1 + X_2 \\ X_2 - X_3 \end{pmatrix}, \quad E = \text{diag}(\varepsilon_i), \quad D = \begin{pmatrix} 0 \\ E \end{pmatrix}, \quad R = X_0, \quad \Delta = \frac{E + I}{2}. \quad (2)$$

In the notation of [2, 3], our matrices (C, D, R, E) are $(\mathcal{A}, \mathcal{B}, \mathcal{X}_0, \mathcal{E})$. In [1], they are $(\mathcal{A}, \mathcal{B}\mathcal{E}, \mathcal{X}_0, \mathcal{E})$: that paper defines $\mathcal{B} = \begin{pmatrix} 0 \\ I \end{pmatrix}$, whereas here $D = \begin{pmatrix} 0 \\ E \end{pmatrix}$. Our A, B below agree with the gluing matrices $\mathbf{A}, \mathbf{B}$ in these sources. Each row of X_k has one entry equal to 1, in the column of the corresponding quotient face. Distinct local faces can give the same column. Local orders, rather than a global ordering of ideal vertices, are the data being fixed.

Use the shape placement of those papers: z corresponds to edges 01, 23; for a positive tetrahedron, z' corresponds to 03, 12 and z'' to 02, 13. These latter two pairs exchange for a negative tetrahedron. Delete one redundant edge equation and append the equation for a fixed preferred longitude λ . With the resulting shape exponent matrices G, G', G'' , put

$$A = G - G', \quad B = G'' - G'. \quad (3)$$

Lemma 1.1 (Rank and symplectic identities). *The edge rows have rank $N - 1$, and their sole linear relation is their sum. After one edge row is deleted and the row for any nonzero class in $H_1(\partial M; \mathbb{Q})$ is appended, the resulting matrices satisfy*

$$\text{rank}(A \mid B) = N, \quad AB^\top = BA^\top. \quad (4)$$

These facts require no shape solution or angle structure.

Proof. Apply the combinatorial complex in [5, Theorem 4.1] to the ideal compactification. Over $\mathbb{Q}$, its edge map has kernel generated by the sum of the edges when there is one ideal vertex. Thus its image has dimension $N - 1$. The theorem makes this image isotropic in the $2N$ -dimensional space of tetrahedral edge-pair coordinates. The signed-corner map of [5, Lemma 4.3] injects $H_1(\partial M; \mathbb{Q})$ into the symplectic orthogonal of that image, modulo the image itself. Therefore a nonzero peripheral class adds an independent row orthogonal to all edge rows. Its pairing with itself is zero because the pairing is alternating. Eliminating the prime coordinate in each tetrahedron writes these rows as $(A \mid B)$; their isotropy is precisely the second identity in (4). $\square$

The preferred longitude is nonzero in $H_1(\partial M; \mathbb{Q})$, as required by this lemma, but maps to zero in $H_1(M; \mathbb{Q})$. Its vanishing in the interior homology is a different fact, used in Section 5. Our homogeneous shape coordinates are

$$(\delta \log z, \delta \log z', \delta \log z'') = (t, s - t, -s), \quad (5)$$

so their gluing relation is $At - Bs = 0$. The peripheral row is the unhalved sum of signed corner contributions around λ , as in [1, 3]. Dividing this row by two would change the integral normalization

and the determinant statement below. The variables in (5) are formal homogeneous logarithmic coordinates. Their use does not assume that a geometric shape solution has been chosen.

We use $X = M/\partial M$ for the ideal compactification. Unless a coefficient ring is displayed, kernels, images, ranks and nullities are over $\mathbb{Q}$, or over $\mathbb{R}$ after extension of scalars; the ranks are the same. For an integer matrix T with m rows, $\text{coker } T$ means $\mathbb{Z}^m/T\mathbb{Z}^{\text{cols}(T)}$. A sublattice is *primitive* if its quotient in the ambient lattice is torsion-free. The bars in $\overline{C}$ and $\overline{R}$ denote reversal of the local orders, not complex conjugation.

Symbol	Meaning and size
$C, \overline{C}$	Original and reversed face matrices, $2N \times 2N$
D	Signed input matrix for face constraints, $2N \times N$
$R, \overline{R}$	Face selectors X_0, X_3 , of size $N \times 2N$
E, Δ	$\text{diag}(\varepsilon_i)$ and $\text{diag}((1 + \varepsilon_i)/2)$
A, B	Edge and preferred-longitude gluing matrices, $N \times N$
n	nullity C
W	Common solvability space in Section 3

1.2 Main statements

Theorem 1.2 (Rational face–edge duality). *With these conventions, put $n = \text{nullity } C$. Then*

1. *nullity $B = 2n$.*
2. *If C is invertible, then B is invertible and*

$$B^{-1}A = RC^{-1}D + (RC^{-1}D)^{\top} + \Delta. \quad (6)$$

3. *Without invertibility, the face constraint space and its symmetric form coincide with the corresponding Neumann–Zagier relation, as specified in Section 7.*

No angle structure is needed for these rational assertions.

Theorem 1.3 (Integral kernel and cokernel duality). *Let $\overline{C}$ denote the face matrix with all vertex orders reversed, and set $\overline{R} = X_3$. With integer kernels and cokernels, one has*

$$\ker_{\mathbb{Z}} B = R \ker_{\mathbb{Z}} C \oplus \overline{R} \ker_{\mathbb{Z}} \overline{C}. \quad (7)$$

There is a short exact sequence

$$0 \longrightarrow \text{coker } C \oplus \text{coker } \overline{C} \xrightarrow{\iota} \text{coker } B \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0, \quad (8)$$

where $\iota([Dt]_C, [Dt]_{\overline{C}}) = [At]_B$. Every element of the domain admits the indicated representation. The two face matrices have identical Smith normal forms. No splitting of (8) is asserted.

Theorem 1.4 (Determinant relation). *With the same hypotheses and the unhalved preferred-longitude convention,*

$$|\det B| = 2|\det C|^2. \quad (9)$$

For geometric triangulations, parts (1) and (2) of Theorem 1.2 give parts (1) and (3) of Conjecture 4.1 in [1]. The determinant magnitudes asserted in part (2) of that conjecture are separate. The strict rank inequality needed for the following consequence is proved in Section 8.

Theorem 1.5 (Generalized FAMED). *Every ordered ideal triangulation of a hyperbolic knot complement in S^3 admitting a strict angle structure is generalized FAMED with respect to its preferred longitude, in the sense of Definition 1.1 of [3].*

This is Conjecture 1.2 of [3]. It does not assert the extra longitude–meridian conditions of Definition 1.5 there.

Theorem 1.6 (Cyclic face torsion). *The torsion subgroup of $\text{coker } C$ is cyclic. If C is nonsingular, $\text{coker } C \cong \mathbb{Z}/d\mathbb{Z}$ with $d = |\det C|$. Section 12 realizes d as the divisibility of a filling-core class in the first homology of an explicitly constructed embedded complex. These assertions also hold for an ordered ideal triangulation of an integral homology solid torus.*

1.3 Relation to earlier work and proof structure

The map $x \mapsto Rx$ in (7) gives a kernel correspondence of the kind requested in Wong’s Problem 2.1 of the 2026 RIMS problem collection [7], in the preferred-longitude knot setting. Section 9 gives a face-variable formulation relevant to Problem 2.3 for each fixed lifted logarithmic system. The construction uses gluing matrices to choose an affine origin; it is not a presentation obtained from the face-pairing data alone. No existence of geometric solutions follows from this change of variables.

The established inputs are the Neumann–Zagier symplectic and rank results [4, 5], the integral normal-surface lattice theorem of Garoufalidis, Hodgson, Hoffman and Rubinstein [6], and Wong’s analytic theorem [3]. The contribution proposed here is the explicit correspondence between their gluing data and the face matrix C , together with its integral and topological consequences. The reversal invariance discussed in Section 3.3 of [1] is refined to an equality of Smith normal forms.

The proof is organized as follows. Sections 2–3 construct the face relation and compute its dimension. Sections 4–5 identify it with the full edge-and-longitude relation. Section 6 proves uniqueness of the face lift, and Section 7 extracts the constrained symmetric form when C is singular. Section 8 proves positive gluing rank under strict angles and applies Wong’s theorem. The affine formulation is followed by integral reversal, lattice, and embedded-complex arguments in Sections 10–12.

Example 1.7 (The figure-eight knot and the peripheral choice). Use the two-tetrahedron triangulation and the face labels of Figure 1 in [7]. Number its four quotient faces from 0 to 3. The local face lists are $(1, 3, 2, 0)$ and $(2, 0, 1, 3)$, and the tetrahedron signs are $+1, -1$. In our notation,

$$C = \begin{pmatrix} 0 & 1 & 1 & -1 \\ -1 & 1 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad R = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

The retained edge equation and the preferred longitude give

$$A = \begin{pmatrix} 1 & -2 \\ 0 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & -1 \\ 0 & 2 \end{pmatrix}.$$

Consequently $RC^{-1}D = \text{diag}(-1, 1)$ and $\Delta = \text{diag}(1, 0)$, so both sides of (6) equal $\text{diag}(-1, 2)$. The homogeneous relation is $s_1 = -t_1$, $s_2 = 2t_2$. Also $\det C = -1$ and $\det B = -2$, in agreement with (9).

Replacing the longitude by the meridian changes B to

$$B_\mu = \begin{pmatrix} -1 & -1 \\ 1 & 1 \end{pmatrix}.$$

This matrix is singular although C is invertible. Thus the preferred-longitude hypothesis cannot be replaced by an arbitrary peripheral curve. The affine constants for the same example are computed in Section 9.

2 The pairing and reversal

The proof uses two face systems, one for the chosen order and one for its reverse. Their interaction is encoded by a bilinear pairing on local face values. All vector spaces in Sections 2–8 may be taken over $\mathbb{Q}$; we use $\mathbb{R}$ to keep the orthogonality notation familiar.

Reverse every local order, retaining the quotient-face and tetrahedron labels. The permutation $0123 \mapsto 3210$ is even, so it preserves the tetrahedron sign. The reversed face matrix and restriction map are

$$\overline{C} = \begin{pmatrix} X_3 - X_2 + X_1 \\ X_1 - X_0 \end{pmatrix}, \quad \overline{R} = X_3.$$

Let $V = \mathbb{R}^{4N}$ denote the space of local face assignments. Equip it with the nondegenerate symmetric pairing

$$\langle x, y \rangle = \sum_i \varepsilon_i (x_{i0}y_{i0} - x_{i1}y_{i1} + x_{i2}y_{i2} - x_{i3}y_{i3}). \quad (10)$$

The subspace $F \subset V$ of assignments constant on paired faces has dimension $2N$. Equation (1) cancels the two contributions of each quotient face, so $\langle F, F \rangle = 0$ and hence $F = F^\perp$.

The local homogeneous constraint planes are

$$L_i = \{(u, u + v, v, v) : u, v \in \mathbb{R}\}, \quad \overline{L}_i = \{(p, p, p + w, w) : p, w \in \mathbb{R}\}.$$

Their pairing is zero, and both have dimension two. Thus $\overline{L}_i = L_i^\perp$. Set $L = \bigoplus_i L_i$. Under the natural identification of quotient-face assignments with F , the kernels of C and $\overline{C}$ are $F \cap L$ and $F \cap L^\perp$. Consequently

$$\dim(F \cap L^\perp) = \dim(F + L)^\perp = 4N - \dim(F + L) = \dim(F \cap L),$$

and the reversed matrix has the same nullity n as C .

We also need an identity for the inhomogeneous systems. If $Cx = Dt$ and $\overline{C}y = Dt'$, their local values have the form

$$x_i = (u_i, u_i + v_i, v_i, v_i - \varepsilon_i t_i), \quad y_i = (p_i - \varepsilon_i t'_i, p_i, p_i + w_i, w_i).$$

Since $x, y \in F$, their pairing is zero. Its i th summand is $-t'_i u_i + t_i w_i$, giving

$$(t')^\top R x = t^\top \overline{R} y. \quad (11)$$

In particular, if C is invertible then so is $\overline{C}$, and

$$\overline{R} \overline{C}^{-1} D = (R C^{-1} D)^\top. \quad (12)$$

3 The face relation

We now allow singular face matrices. Define

$$\begin{aligned} S_C &= \{t : Dt \in \text{im } C\}, & O_C &= (R \ker C)^\perp, \\ S_{\overline{C}} &= \{t : Dt \in \text{im } \overline{C}\}, & O_{\overline{C}} &= (\overline{R} \ker \overline{C})^\perp. \end{aligned}$$

Here perpendiculars in $\mathbb{R}^N$ use the ordinary dot product, whereas perpendiculars in V use (10).

Lemma 3.1. *One has $S_{\overline{C}} = O_C$ and $S_C = O_{\overline{C}}$.*

Proof. A local particular solution of the reversed equations is $y_i^0 = (-\varepsilon_i t_i, 0, 0, 0)$. There is a face-constant solution precisely when the affine space $y^0 + L^\perp$ meets F . This is equivalent to $y^0 \in F + L^\perp$, and therefore to $y^0 \perp (F + L^\perp)^\perp = F \cap L$. For $k_i = (u_i, u_i + v_i, v_i, v_i)$, the corresponding local pairing is $-t_i u_i$. Thus the solvability condition is exactly $t \perp R \ker C$. The original particular solution $x_i^0 = (0, 0, 0, -\varepsilon_i t_i)$ gives the second equality in the same way: its pairing with $(p_i, p_i, p_i + w_i, w_i)$ is $t_i w_i$. $\square$

Put $W = S_C \cap S_{\overline{C}} = S_C \cap O_C$, and define a linear relation between two copies of $\mathbb{R}^N$ by

$$\mathcal{L}_F = \{(t, s) : Cx = Dt, \overline{C}y = Dt, s = Rx + \overline{R}y + \Delta t \text{ for some } x, y\}. \quad (13)$$

The projection $(t, s) \mapsto t$ has image W . Its kernel within $\mathcal{L}_F$ is $\{0\} \times (R \ker C + \overline{R} \ker \overline{C})$. By Lemma 3.1,

$$R \ker C + \overline{R} \ker \overline{C} = (O_C \cap O_{\overline{C}})^\perp = W^\perp.$$

The projection therefore gives a short exact sequence of vector spaces

$$0 \longrightarrow W^\perp \xrightarrow{s \mapsto (0, s)} \mathcal{L}_F \xrightarrow{(t, s) \mapsto t} W \longrightarrow 0,$$

and in particular

$$\dim \mathcal{L}_F = \dim W^\perp + \dim W = N. \quad (14)$$

At this stage no injectivity of the face parametrization has been used.

4 Local cochains and the edge equations

Fix x, y, t, s as in (13). For each quotient face f , write its ordered vertices as $a < b < c$ and assign potentials

$$\phi_f(a) = 0, \quad \phi_f(b) = x_f, \quad \phi_f(c) = x_f - y_f.$$

Their differences define an additive cochain on the three directed edges of the face:

$$h_f(ab) = x_f, \quad h_f(ac) = x_f - y_f, \quad h_f(bc) = -y_f.$$

The potentials, and therefore the cochains, agree at corresponding vertices and edges of a paired face because its gluing preserves the vertex order. We do not yet assert agreement between two different faces meeting the same tetrahedron edge. Figure 1 records the local convention.

For a directed edge jm of one tetrahedron, let k, l be the two remaining vertices. Denote the cochain on face k by h_k and set

$$q_{jm} = \varepsilon \operatorname{sgn}(j, m, k, l)(h_k(jm) - h_l(jm)). \quad (15)$$

Exchanging j, m changes the permutation sign and both cochain signs; exchanging k, l changes the permutation sign and the order of the difference. Thus q_{jm} is an unoriented edge value, independent of both choices. The local constraints are

$$x_1 = x_0 + x_2, \quad x_2 - x_3 = \varepsilon t, \quad y_2 = y_1 + y_3, \quad y_1 - y_0 = \varepsilon t.$$

They give

$$\begin{aligned} q_{01} &= q_{23} = t, \\ q_{03} &= q_{12} = \varepsilon(x_0 + y_3), \\ q_{02} &= q_{13} = -t - \varepsilon(x_0 + y_3). \end{aligned} \quad (16)$$

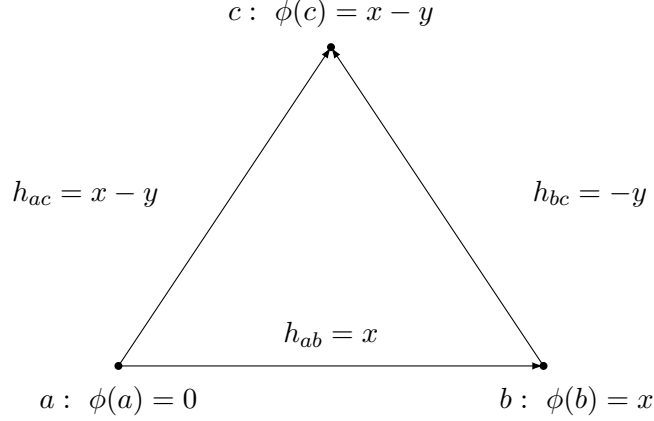


Figure 1: The local cochain on an ordered face with vertices $a < b < c$. Each arrow is oriented from the lower to the higher vertex. Its value is the difference of the endpoint potentials, so $h_{ab} + h_{bc} = h_{ac}$.

For example, $q_{01} = \varepsilon(x_2 - x_3) = t$ and $q_{03} = \varepsilon[(x_1 - y_1) - (x_2 - y_2)] = \varepsilon(x_0 + y_3)$. Since $s = x_0 + y_3 + (\varepsilon + 1)t/2$, the positive-tetrahedron placement gives $(q_z, q_{z'}, q_{z''}) = (t, s - t, -s)$. When $\varepsilon = -1$, exchanging the prime and double-prime edge pairs gives the same triple. This is the placement fixed in (5).

For a fixed directed quotient edge, each local term in (15) is the incidence of an adjacent face times its cochain value. Along a paired face, the two tetrahedra induce opposite face orientations by (1), while the directed-edge cochain value agrees. The two contributions therefore cancel. Summing over the complete link of a quotient edge gives zero. This counts local incidences separately and applies equally to self-gluing and repeated edges. Hence the values in (16) satisfy every homogeneous edge equation.

5 The longitude via a dual-spine cocycle

The edge cancellation does not by itself establish the peripheral equation. To obtain that equation, we construct a cohomology class of M whose periods are the peripheral shape holonomies of the face data. Its period on the preferred longitude is zero because that curve is null-homologous in M .

Write $\phi_k(j)$ for the potential at vertex j on local face k . In each tetrahedron define four auxiliary numbers

$$\begin{aligned} f_0 &= 0, & f_1 &= -x_2 - x_3, \\ f_2 &= x_0 - x_2 - x_3 + y_3, & f_3 &= x_0 - 2x_3 + y_3. \end{aligned} \tag{17}$$

If faces k, l meet in edge jm , then

$$\phi_k(j) + \phi_k(m) - \phi_l(j) - \phi_l(m) = f_l - f_k. \tag{18}$$

Here is the full local check, with the pairs $k < l$ in increasing order:

(k, l)	$\{j, m\}$	$f_l - f_k$
$(0, 1)$	$\{2, 3\}$	$-x_2 - x_3$
$(0, 2)$	$\{1, 3\}$	$x_0 - x_2 - x_3 + y_3$
$(0, 3)$	$\{1, 2\}$	$x_0 - 2x_3 + y_3$
$(1, 2)$	$\{0, 3\}$	$x_0 + y_3$
$(1, 3)$	$\{0, 2\}$	$x_0 + y_3 + \varepsilon t$
$(2, 3)$	$\{0, 1\}$	εt

Substitution of the face potentials into the left side of (18), followed by the four local constraints from Section 4, gives these entries.

The cusp sign. Consider a normal arc in the cusp triangle at vertex j . Suppose it enters face k , leaves face l , and cuts off the corner corresponding to edge jm . The truncation triangle has induced boundary orientation $-\varepsilon(-1)^j$ relative to the increasing order of its three vertices: it is parallel to the face opposite j , with the opposite outward normal. The arc goes from side (m, l) to side (m, k) . Its signed turn is therefore

$$[-\varepsilon(-1)^j] \operatorname{sgn}(m, l, k \text{ relative to the remaining increasing triple}) = \varepsilon \operatorname{sgn}(j, m, k, l).$$

Multiplying this sign by q_{jm} in (15) gives the signed shape contribution of the arc. Equation (18) rewrites that contribution as

$$h_k(jm) - h_l(jm) = [f_l + 2\phi_l(j)] - [f_k + 2\phi_k(j)]. \quad (19)$$

If the peripheral holonomy convention uses the opposite global turning sign, all these periods change sign together; their vanishing is unchanged.

Closure on the spine. The dual two-skeleton of the ideal triangulation is a spine of M . On a directed dual edge crossing from local face (i, k) to its paired face (i', l) , put

$$a((i, k) \longrightarrow (i', l)) = f_{i,k} - f_{i',l}.$$

The two local face occurrences remain distinct when $i = i'$, so this also defines a on a dual loop. Reversing the dual edge negates its value.

To prove that a is closed, traverse the boundary of a dual two-cell, which encircles one quotient edge. In successive tetrahedral wedges, let k_r and l_r be the incoming and outgoing faces. The period of a around the cell is

$$\sum_r (f_{i_r, l_r} - f_{i_r, k_r}).$$

By (18), each term is the incoming sum of endpoint potentials minus the outgoing sum. The outgoing face of one wedge is paired with the incoming face of the next, and the two endpoint potentials agree. Every term cancels. The argument counts occurrences, so it also applies to loop edges and self-gluing. Therefore a represents a class in $H^1(M; \mathbb{R})$.

Peripheral periods. On a closed normal peripheral curve, the ϕ terms in (19) cancel across successive paired faces. Its shape holonomy is consequently $\sum_r (f_{i_r, l_r} - f_{i_r, k_r})$. Pushing the curve into the dual spine follows exactly those face crossings, so this sum is its period under a . The push preserves the homotopy class: within each truncated tetrahedron one replaces the local arc by a path through its center and the centers of the crossed faces. Since $[\lambda] = 0$ in $H_1(M; \mathbb{R})$, the longitude period is zero.

Combined with the edge equations from Section 4, this proves $\mathcal{L}_F \subseteq \ker(A \mid -B)$. Lemma 1.1 gives dimension N for the right side, and (14) gives the same dimension for the left side. Thus

$$\mathcal{L}_F = \ker(A \mid -B). \quad (20)$$

The block-column matrix $\begin{pmatrix} B^\top \\ A^\top \end{pmatrix}$ has rank N and, by $AB^\top = BA^\top$, its image lies in this kernel. Equality of dimensions gives

$$\ker(A \mid -B) = \operatorname{im} \begin{pmatrix} B^\top \\ A^\top \end{pmatrix}, \quad W = \operatorname{im} B^\top.$$

6 Unique face lifts and the nonsingular identity

We have identified the face and edge relations without assuming that face lifts are unique. Their uniqueness is the topological step responsible for nullity doubling.

Consider the map

$$P : \ker C \oplus \ker \overline{C} \longrightarrow \mathbb{R}^N, \quad P(x, y) = Rx + \overline{R}y.$$

Suppose $P(x, y) = 0$. Then $t = s = 0$ in (13), so every q_{jm} in (16) is zero. Equation (15) says that the cochains on the two faces meeting each tetrahedron edge agree along that edge. They also agree across paired faces. Following the connected link of each quotient edge gives a well-defined cochain on the edges of the ideal compactification $X = M/\partial M$. On every triangle it is a difference of vertex potentials, so it is a cellular 1-cocycle.

Conversely, let $h \in Z^1(X; \mathbb{R})$. On a face with increasing vertices $a < b < c$, put $x_f = h(ab)$ and $y_f = -h(bc)$. These are quotient-face values, since h is global and the gluing preserves the order. In a tetrahedron they read

$$\begin{aligned} (x_0, x_1, x_2, x_3) &= (h_{12}, h_{02}, h_{01}, h_{01}), \\ (y_0, y_1, y_2, y_3) &= (-h_{23}, -h_{23}, -h_{13}, -h_{12}). \end{aligned}$$

The triangle equations $h_{ab} + h_{bc} = h_{ac}$ give $Cx = \overline{C}y = 0$, and $Rx + \overline{R}y = 0$ follows directly. The reconstructed face potentials recover h because $x_f - y_f = h(ac)$. These two constructions are inverse, hence

$$\ker P \cong Z^1(X; \mathbb{R}).$$

The complex X has one vertex, so the coboundary $C^0(X; \mathbb{R}) \rightarrow C^1(X; \mathbb{R})$ is zero. Therefore $Z^1(X; \mathbb{R}) = H^1(X; \mathbb{R})$. The quotient description and Poincaré–Lefschetz duality give

$$H^1(X; \mathbb{R}) \cong H^1(M, \partial M; \mathbb{R}) \cong H_2(M; \mathbb{R}) = 0.$$

The last equality holds for a knot exterior in S^3 . Thus P is injective. Its domain has dimension $2n$, and its image is $W^\perp$ by Section 3. Since $W = \operatorname{im} B^\top$,

$$\text{nullity } B = \dim W^\perp = 2n.$$

This proves the first part of Theorem 1.2. It also proves uniqueness of every face lift in (20), because the difference of two lifts belongs to $\ker P$.

If $n = 0$, the face relation is the graph of

$$s = (RC^{-1}D + \overline{RC}^{-1}D + \Delta)t.$$

Nullity doubling makes B invertible. Comparing this graph with (20) and using (12) gives

$$B^{-1}A = RC^{-1}D + (RC^{-1}D)^\top + \Delta,$$

which proves the second part of Theorem 1.2.

7 Singular constraint spaces and forms

Choose invertible rational matrices E_1, E_2 such that

$$E_1CE_2 = \text{diag}(I_r, 0_n), \quad r = 2N - n.$$

For a matrix partitioned accordingly, a subscript r denotes the first r columns or rows and a subscript n the remaining ones. Define

$$K_F = \begin{pmatrix} (RE_2)_n^\top \\ (E_1D)_n \end{pmatrix},$$

$$H_F = (RE_2)_r(E_1D)_r, \quad G_F = H_F + H_F^\top + \Delta.$$

The last n columns of E_2 form a basis of $\ker C$, so the first block of K_F imposes $t \perp R \ker C$. The second block imposes solvability of $Cx = Dt$. Thus

$$\ker K_F = W, \quad \text{rank } K_F = 2n.$$

The rank follows from $\text{codim } W = 2n$, already proved; there is no assumption that these two sets of constraints are independent before that argument.

For $t \in W$, a particular solution is $x_t = (E_2)_r(E_1D)_rt$. If $(t, s) \in \mathcal{L}_F$, choose a reversed solution $\overline{C}y = Dt$. For $v \in W$, changes of an original solution lie in $\ker C$ and disappear after multiplication by $v^\top R$. Applying (11) to $Cx_v = Dv$ and $\overline{C}y = Dt$ gives

$$v^\top \overline{R}y = t^\top Rx_v.$$

Consequently

$$v^\top s = v^\top Rx_t + t^\top Rx_v + v^\top \Delta t = v^\top G_F t.$$

Conversely, for fixed $t \in W$ every s in the fiber differs from any one such value by $W^\perp$. Hence the entire relation, including its off-diagonal bilinear information, is

$$\mathcal{L}_F = \{(t, s) : t \in W, s - G_F t \in W^\perp\}.$$

To compare this with the row conditions in [3, Definition 1.1], row-reduce $(B \mid A)$ and write its N independent rows as

$$\begin{pmatrix} B_0 & A_0 \\ 0 & K_{NZ} \end{pmatrix},$$

where B_0 has $N - 2n$ rows and K_{NZ} has $2n$ rows. The equation $At = Bs$ is equivalent to $K_{NZ}t = 0$ and $A_0t = B_0s$. Since B_0 has full row rank, the second equation can be solved for s for every t satisfying the first. Equation (20) therefore gives

$$\ker K_{NZ} = W = \ker K_F.$$

The matrices K_{NZ} and K_F both have $2n$ independent rows, so they are row equivalent.

Furthermore, $\text{row } B_0 = \text{row } B = W$. Thus B_0 annihilates $W^\perp$. The preceding description of $\mathcal{L}_F$ now gives $A_0t = B_0G_Ft$ for every $t \in W$. Each row of $A_0 - B_0G_F$ lies in $W^\perp = \text{row } K_{NZ}$. There is consequently a rational matrix Q with $A_0 - B_0G_F = QK_{NZ}$, and the explicit invertible row operation

$$\begin{pmatrix} I & -Q \\ 0 & I \end{pmatrix} \begin{pmatrix} B_0 & A_0 \\ 0 & K_{NZ} \end{pmatrix} = \begin{pmatrix} B_0 & B_0G_F \\ 0 & K_{NZ} \end{pmatrix}$$

proves the required quadratic compatibility. These arguments establish the third part of Theorem 1.2 for every rational choice of E_1, E_2 .

8 Strict angles and positive gluing rank

A strict angle structure assigns an angle in $(0, \pi)$ to each pair of opposite tetrahedron edges, with sum π in each tetrahedron and sum 2π around each quotient edge. In particular, each quotient edge has degree at least three, where degree counts local edge occurrences. The following combinatorial lemma provides the strict rank inequality that does not follow from nullity doubling alone.

Lemma 8.1. *If the ordered knot triangulation admits a strict angle structure, its full edge block B_{edge} is nonzero. Consequently $\text{rank } B > 0$.*

Proof. Suppose instead that $B_{\text{edge}} = 0$. Orient local edges and faces by increasing vertex order; these orientations agree across paired faces. In the cellular chain complex of $X = M/\partial M$, write e_{ab} for the quotient-edge generator represented by local edge ab with $a < b$, and f_k for a local face viewed as a quotient-face generator. The local boundaries are

$$\begin{aligned} \partial_2 f_0 &= e_{12} - e_{13} + e_{23}, & \partial_2 f_1 &= e_{02} - e_{03} + e_{23}, \\ \partial_2 f_2 &= e_{01} - e_{03} + e_{13}, & \partial_2 f_3 &= e_{01} - e_{02} + e_{12}. \end{aligned}$$

A zero tetrahedron column in B_{edge} means

$$e_{03} + e_{12} = e_{02} + e_{13} \tag{21}$$

in the free vector space on quotient edges. The tetrahedron sign only exchanges the two sides. Thus both $f_0 - f_1$ and $f_2 - f_3$ belong to $\ker \partial_2$ for each tetrahedron. Since $H_1(X; \mathbb{Q}) = 0$ and X has one vertex, N edges and $2N$ faces,

$$\text{rank } \partial_2 = N, \quad \dim \ker \partial_2 = N.$$

Construct a multigraph whose vertices are the $2N$ quotient faces. Tetrahedron i contributes an edge P_i joining f_{i0} to f_{i1} and an edge Q_i joining f_{i2} to f_{i3} . Every quotient face has two local occurrences, each used once, so the graph is 2-regular, counting a loop twice. If it has k components, its incidence vectors span a space of dimension $2N - k$. These vectors are the displayed face differences in $\ker \partial_2$, so $2N - k \leq N$, or $k \geq N$.

A loop P_i glues faces 0 and 1 of tetrahedron i to each other. The ordered gluing fixes their common edge 23 pointwise. Both sides of that local edge link are then paired to each other, giving a quotient edge of degree one. A loop Q_i similarly gives degree one at edge 01. Strict angles exclude these possibilities. Every graph component therefore has at least two vertices. Together with $k \geq N$, this forces exactly N components, each with two vertices and two parallel edges.

If a component consists of P_i, P_j , the two face gluings pair faces 0,1 of tetrahedron i with faces 0,1 of tetrahedron j . Their common edges $23_i, 23_j$ form a closed link cycle of length two: both adjacent faces of both edge occurrences are already paired within the component. Hence the resulting quotient edge has degree two. A component of two Q edges has the same property at edge 01. Strict angles exclude these cases as well. Every component must therefore consist of one P_i and one Q_j . In particular, each tetrahedron appears exactly once as an index i and once as an index j among these mixed components.

There are two possible ordered face pairings in a mixed component. For $f_{i0} \sim f_{j2}$ and $f_{i1} \sim f_{j3}$, the induced edge identifications are

$$\begin{aligned} e_{12}(i) &= e_{01}(j), & e_{13}(i) &= e_{03}(j), & e_{23}(i) &= e_{13}(j), \\ e_{02}(i) &= e_{01}(j), & e_{03}(i) &= e_{02}(j), & e_{23}(i) &= e_{12}(j). \end{aligned}$$

For $f_{i0} \sim f_{j3}$ and $f_{i1} \sim f_{j2}$, they are

$$\begin{aligned} e_{12}(i) &= e_{01}(j), & e_{13}(i) &= e_{02}(j), & e_{23}(i) &= e_{12}(j), \\ e_{02}(i) &= e_{01}(j), & e_{03}(i) &= e_{03}(j), & e_{23}(i) &= e_{13}(j). \end{aligned}$$

Both tables imply $e_{12}(i) = e_{02}(i)$ and $e_{12}(j) = e_{13}(j)$. Since every tetrahedron occurs in each role, $e_{02} = e_{12} = e_{13}$ in every tetrahedron. Equation (21) then forces e_{03} to be the same quotient edge. This is equality of basis vectors in $C_1(X; \mathbb{Q})$, not merely equality of homology classes. Denote the common edge by c_i .

Either pairing table gives $c_i = c_j$, $e_{01}(j) = c_i$, and $e_{23}(i) = c_j$. Every face gluing belongs to a mixed component. The tetrahedron adjacency graph is connected, so all local edges in the triangulation represent a single quotient edge. Since their number is N , it follows that $N = 1$.

For one tetrahedron, (1) allows only opposite-parity faces to be paired. The two possibilities are (01)(23) and (03)(12). The first glues faces 0,1 together, already shown to give degree one. In the second, the ordered gluing of face 1 = (023) to face 2 = (013) fixes edge 03 pointwise, again giving degree one. This contradicts strict angles and proves $B_{\text{edge}} \neq 0$.

Finally, the sum of all edge rows of B_{edge} is zero: each tetrahedron has two occurrences of each shape type. If deleting one row left a zero matrix, that row would be zero as well. Thus the retained edge rows, and hence the longitude matrix B , are nonzero. $\square$

For clarity, in our notation [3, Definition 1.1] requires the following four conditions for generalized FAMED with respect to the longitude:

1. the strict angle space is nonempty;
2. nullity $B = 2n < N$, where $n = \text{nullity } C$;
3. K_{NZ} and K_F from Section 7 are row equivalent;
4. the two block matrices

$$\begin{pmatrix} B_0 & A_0 \\ 0 & K_{NZ} \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} B_0 & B_0 G_F \\ 0 & K_{NZ} \end{pmatrix}$$

are row equivalent.

Here (B_0, A_0, K_{NZ}) is obtained from the reduced row echelon form of $(B \mid A)$, as in that definition, and our G_F is its matrix $\mathcal{G}$.

Proof of Theorem 1.5. The first condition is the angle-structure hypothesis. Theorem 1.2 and Lemma 8.1 give

$$\text{nullity } B = 2n = N - \text{rank } B < N,$$

which is the second condition. The two row equivalences were proved in Section 7, including an explicit row operation for the fourth condition. $\square$

The rank lemma uses only the absence of degree-one and degree-two edges, together with $H_1(X; \mathbb{Q}) = 0$. It asserts that the full edge block is nonzero; it does not assert that every tetrahedron column is nonzero.

Corollary 8.2 (Wong’s upper bound without a FAMED hypothesis). *Let $\mathcal{AT}$ be the nonempty strict angle space of an ordered ideal triangulation of a hyperbolic knot complement in S^3 . For $\alpha \in \mathcal{AT}$, let $\mathcal{Z}_h(\mathcal{T}, \alpha)$ be its Teichmüller TQFT partition function, and let $H_\lambda^\mathbb{R}$ denote angular longitude holonomy, with the conventions of [3]. Then*

$$\limsup_{h \rightarrow 0^+} 2\pi h \log |\mathcal{Z}_h(\mathcal{T}, \alpha)| \leq - \sup_{\substack{\alpha' \in \mathcal{AT} \\ H_\lambda^\mathbb{R}(\alpha') = H_\lambda^\mathbb{R}(\alpha)}} \text{Vol}(\alpha').$$

Here Vol is the sum of the ideal-tetrahedron volumes determined by the angles.

Proof. Theorem 1.5 supplies the generalized FAMED hypothesis in the first clause of [3, Theorem 1.11]. That analytic theorem gives the inequality. No further estimate is needed. $\square$

Corollary 8.3 (The partition-function volume limit). *Suppose $\mathcal{T}$ is an ordered complete geometric ideal triangulation of a hyperbolic knot complement $M = S^3 \setminus K$, and let α_{geom} be its geometric angle structure. For every strict angle structure α with $H_\lambda^\mathbb{R}(\alpha) = H_\lambda^\mathbb{R}(\alpha_{\text{geom}})$, one has*

$$\lim_{h \rightarrow 0^+} 2\pi h \log |\mathcal{Z}_h(\mathcal{T}, \alpha)| = -\text{Vol}(M).$$

In particular, the conclusion holds for $\alpha = \alpha_{\text{geom}}$. It allows singular face matrices.

Proof. Theorem 1.5 supplies generalized FAMED. Let $\mathbf{z}_{\text{geom}}$ denote the complete shapes, all with positive imaginary parts. They satisfy the edge equations on the geometric logarithmic branch. Completeness makes the real part of their logarithmic longitude holonomy zero, and the imaginary part is the angular holonomy of [3, Section 2.2]. Any constants introduced by simplifying the corner expressions are retained in both holonomies. Thus

$$H_\lambda^\mathbb{C}(\mathbf{z}_{\text{geom}}) = iH_\lambda^\mathbb{R}(\alpha_{\text{geom}}) = iH_\lambda^\mathbb{R}(\alpha).$$

The additional shape-solution hypothesis of the second clause of [3, Theorem 1.11] is therefore satisfied. Its logarithmic limit gives the conclusion, since the sum of these tetrahedral volumes is $\text{Vol}(M)$. $\square$

The angular holonomies in this corollary use the same normal peripheral representative and logarithmic convention. When the lift is normalized to have zero holonomy at the complete structure, its hypothesis reads $H_\lambda^\mathbb{R}(\alpha) = 0$. Exponentiated completeness alone does not select a logarithmic lift.

These consequences use Wong’s analytic theorem. They do not establish the existence of a complete geometric triangulation for every knot, or the additional longitude–meridian conditions in [3, Definition 1.5] needed for its Jones-function assertions.

9 An affine face formulation of the gluing equations

We now express the lifted gluing equations in face coordinates. Related correspondences between logarithmic gluing equations and critical-point equations appear in [2, Proposition 3.4] and [3, Lemma 3.6 and Proposition 3.9]. The construction here uses the face-cochain isomorphism proved above; it does not require a potential function or invertibility of C .

Give each quotient face two complex coordinates x_f, y_f . Define the complex linear space $\mathcal{F}$ by

$$\begin{aligned} (X_0 - X_1 + X_2)x &= 0, \\ (X_3 - X_2 + X_1)y &= 0, \\ (X_2 - X_3)x &= (X_1 - X_0)y. \end{aligned} \tag{22}$$

Set

$$t = E(X_2 - X_3)x, \quad s = Rx + \overline{R}y + \Delta t.$$

These are the simultaneous face equations with t eliminated. By the duality and injectivity arguments in Sections 5–6, the map

$$\Phi : \mathcal{F} \longrightarrow \ker_{\mathbb{C}}(A \mid -B), \quad (x, y) \longmapsto (t, s)$$

is a complex linear isomorphism. Thus $\dim_{\mathbb{C}} \mathcal{F} = N$, including when C is singular.

9.1 The lifted solution set

Fix $u \in \mathbb{C}$ and $\kappa \in \mathbb{Z}^N$. The first $N - 1$ entries of κ are even, and are 2 for the usual geometric edge equations. Fix also the representative and additive constant in the longitude formula: if

$$H_\lambda = i\pi c_\lambda + (GZ + G'Z' + G''Z'')_N, \quad c_\lambda \in \mathbb{Z},$$

then take $\kappa_N = -c_\lambda$, so that the last equation below prescribes $H_\lambda = u$.

Let $\mathcal{S}_{\kappa, u}$ consist of triples $(Z, Z', Z'') \in \mathbb{C}^{3N}$ such that, for every i ,

$$z_i = e^{Z_i} \notin \{0, 1\}, \quad e^{Z'_i} = \frac{1}{1 - z_i}, \quad e^{Z''_i} = 1 - \frac{1}{z_i},$$

and

$$Z + Z' + Z'' = i\pi \mathbf{1}, \quad GZ + G'Z' + G''Z'' = i\pi \kappa + ue_N.$$

Here e_N is the last standard basis vector of $\mathbb{C}^N$. These are normalized logarithmic lifts: the variables are not required to be principal logarithms. No positivity of the shapes or existence of a solution is assumed.

Put $\eta = \kappa - G'\mathbf{1}$ and choose rational vectors satisfying

$$Ap_0 - Br_0 = \eta, \quad Aq - Br = e_N.$$

Such choices exist by the full row rank of $(A \mid B)$. For $(x, y) \in \mathcal{F}$ with $\Phi(x, y) = (t, s)$, define

$$Z = i\pi p_0 + t + uq, \quad S = i\pi r_0 + s + ur. \tag{23}$$

Proposition 9.1. *For each fixed κ and u as above, the subset of $\mathcal{F}$ satisfying*

$$\exp(Z_i)(1 - \exp(-S_i)) = 1, \quad i = 1, \dots, N, \tag{24}$$

is in bijection with $\mathcal{S}_{\kappa, u}$. The bijection sends (x, y) to

$$(Z, Z', Z'') = (Z, i\pi \mathbf{1} - Z + S, -S).$$

Proof. The affine substitutions give $AZ - BS = i\pi\eta + ue_N$. Hence the displayed triple has tetrahedral sum $i\pi\mathbf{1}$ and satisfies

$$GZ + G'Z' + G''Z'' = AZ - BS + i\pi G'\mathbf{1} = i\pi\kappa + ue_N.$$

Equation (24) gives $z_i(1 - z_i'') = 1$ with $z_i = e^{Z_i}$ and $z_i'' = e^{-S_i}$. Both exponentials are nonzero, so $z_i \neq 1$, $z_i'' = 1 - 1/z_i$, and

$$e^{Z'_i} = e^{i\pi - Z_i + S_i} = \frac{1}{1 - z_i}.$$

Thus the triple belongs to $\mathcal{S}_{\kappa, u}$.

Conversely, let $(Z, Z', Z'') \in \mathcal{S}_{\kappa, u}$ and put $S = -Z''$. The tetrahedral sums give $Z' = i\pi\mathbf{1} - Z + S$. Subtracting the two particular solutions in (23) produces

$$(t, s) = (Z - i\pi p_0 - uq, S - i\pi r_0 - ur) \in \ker_{\mathbb{C}}(A \mid -B).$$

The isomorphism Φ gives a unique $(x, y) \in \mathcal{F}$. The shape relation for $e^{Z''}$ gives (24), establishing the inverse. $\square$

9.2 Auxiliary choices and logarithmic lifts

The choices of p_0, r_0, q, r affect only the affine origin. If they are replaced by $\tilde{p}_0, \tilde{r}_0, \tilde{q}, \tilde{r}$ satisfying the same equations, then

$$\delta_u = i\pi(p_0 - \tilde{p}_0, r_0 - \tilde{r}_0) + u(q - \tilde{q}, r - \tilde{r}) \in \ker_{\mathbb{C}}(A \mid -B).$$

Translation by $\Phi^{-1}(\delta_u)$ identifies the two face solution sets and leaves the resulting triple (Z, Z', Z'') unchanged. This assertion keeps κ and u fixed.

The omitted edge equation has forced logarithmic constant

$$i\pi \left(2N - \sum_{j=1}^{N-1} \kappa_j \right),$$

because each tetrahedron contributes twice its logarithmic sum to the sum of all edge equations. It is $2\pi i$ when every retained edge constant is $2\pi i$.

A fixed κ specifies lifted linear constraints, not a unique logarithmic branch. For example, whenever $a, b \in \mathbb{Z}^N$ satisfy $Aa - Bb = 0$, replacing (Z, S) by $(Z + 2\pi ia, S + 2\pi ib)$ preserves κ , u , and all exponentiated shapes. The proposition is a bijection of the specified lifted solution sets; it does not assert a bijection after forgetting logarithmic lifts.

At prescribed u , all unknowns in (22) and (24) are attached to quotient faces. Positive shapes can be selected by imposing $\text{Im}(e^{Z_i}) > 0$; a prescribed principal-logarithm domain requires its corresponding restrictions on Z, Z', Z'' as well. To impose both peripheral completeness equations, substitute (23) into the meridian equation. Setting $u = 0$ alone is not a global assertion of meridian completeness.

9.3 Figure-eight constants

For the ordering in Example 1.7, the gluing data in [1, Figure 4] and [7, Section 2] give

$$A = \begin{pmatrix} 1 & -2 \\ 0 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & -1 \\ 0 & 2 \end{pmatrix}, \quad \kappa = \begin{pmatrix} 2 \\ -2 \end{pmatrix}, \quad \eta = \begin{pmatrix} -1 \\ 2 \end{pmatrix}.$$

The longitude constant here is $c_\lambda = 2$. One may choose

$$p_0 = \begin{pmatrix} 1/3 \\ 1/3 \end{pmatrix}, \quad r_0 = \begin{pmatrix} -1/3 \\ -1/3 \end{pmatrix}, \quad q = \begin{pmatrix} 1/2 \\ 1/4 \end{pmatrix}, \quad r = 0.$$

Direct multiplication gives $Ap_0 - Br_0 = \eta$ and $Aq - Br = e_2$. At $x = y = 0$ and $u = 0$, all three logarithms in each tetrahedron equal $i\pi/3$, recovering the regular shapes $z_i = z'_i = z''_i = e^{i\pi/3}$. This example checks the additive longitude constant as well as the sign of $S = -Z''$.

10 Integral reversal symmetry

Theorem 10.1. *For every ordered oriented face pairing, the integral matrices C and $\overline{C}$ have the same Smith normal form.*

Proof. We first make the local change of coordinates explicit. For four local face values x_0, x_1, x_2, x_3 , write

$$a = x_0 - x_1 + x_2, \quad b = x_2 - x_3, \quad c = x_3 - x_2 + x_1, \quad d = x_1 - x_0.$$

The first pair is the original constraint and the second pair is the reversed constraint. The inverse is integral:

$$x_0 = b + c - d, \quad x_1 = b + c, \quad x_2 = a + d, \quad x_3 = a + d - b.$$

Consequently this change of coordinates is unimodular. Put

$$h = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}, \quad H = \begin{pmatrix} 0 & E \\ E & -E \end{pmatrix}.$$

A direct expansion gives the local bilinear identity

$$\varepsilon \sum_{k=0}^3 (-1)^k x_k x'_k = \varepsilon(a, b) h(a', b')^\top - \varepsilon(c, d) h(c', d')^\top.$$

Thus, after collecting the original and reversed constraints into separate blocks, the global pairing has matrix $\text{diag}(H, -H)$.

Let $\mathcal{F}_\mathbb{Z} \subset \mathbb{Z}^{4N}$ be the lattice of local assignments constant across each paired face. The two diagonal entries of the original pairing corresponding to a paired face are $\sigma, -\sigma$, with $\sigma \in \{1, -1\}$. On this pair, pairing with a constant assignment (r, r) is $\sigma r(u - v)$. Hence the pairing map onto $\mathcal{F}_\mathbb{Z}^*$ is surjective, and its kernel is exactly $\mathcal{F}_\mathbb{Z}$. Applying the unimodular coordinate change gives an exact sequence

$$0 \longrightarrow \mathbb{Z}^{2N} \xrightarrow{\begin{pmatrix} C \\ \overline{C} \end{pmatrix}} \mathbb{Z}^{4N} \xrightarrow{\begin{pmatrix} C^\top H & -\overline{C}^\top H \end{pmatrix}} \mathbb{Z}^{2N} \longrightarrow 0.$$

In particular,

$$C^\top H C = \overline{C}^\top H \overline{C}.$$

Define

$$\Phi : \text{coker } C \longrightarrow \text{coker}(\overline{C}^\top H), \quad [y] \longmapsto [C^\top H y].$$

The displayed identity makes Φ well defined. Surjectivity follows from the last map in the exact sequence: every vector is congruent to a vector $C^\top H y$ modulo $\overline{C}^\top H \mathbb{Z}^{2N}$. If $C^\top H y = \overline{C}^\top H z$, exactness gives $(y, z) = (Cx, \overline{C}x)$ for some integral x , so $[y] = 0$. Thus Φ is an isomorphism. Since H is unimodular and transposition preserves Smith normal form, the theorem follows. $\square$

Reversal preserves each opposite-edge pair and the tetrahedron orientation sign. Retaining the quotient-edge labels and the same normal representative of the peripheral curve therefore retains A and B . Together with (12) and Section 7, this also proves reversal invariance of the constrained face form. Theorem 10.1 strengthens the numerical reversal observation in Section 3.3 of [1] to an equality of integral Smith normal forms.

11 Integral lifts, the cokernel sequence, and determinants

We now prove Theorems 1.3 and 1.4. All kernels and cokernels in this section are taken over $\mathbb{Z}$ unless another coefficient ring is displayed. Put

$$K = \begin{pmatrix} C & 0 & -D \\ 0 & \overline{C} & -D \end{pmatrix}, \quad F_0 = \begin{pmatrix} C & 0 \\ 0 & \overline{C} \\ R & \overline{R} \end{pmatrix}.$$

The ideal compactification $X = M/\partial M$ has one vertex and

$$H_1(X; \mathbb{Z}) \cong H_1(M, \partial M; \mathbb{Z}) \cong H^2(M; \mathbb{Z}) = 0.$$

Here the quotient identification uses connected boundary, the second isomorphism is Poincaré–Lefschetz duality, and the last equality follows from the integral homology of a knot exterior.

11.1 The cokernel of the simultaneous constraints

Lemma 11.1. *There is an integral presentation*

$$\text{coker } K \cong H_1(X; \mathbb{Z}). \quad (25)$$

Proof. Use four row generators p_i, q_i, p'_i, q'_i for each tetrahedron. The last N columns of K impose $q'_i = -q_i$. Make the unimodular substitutions $p_i = \varepsilon_i a_i$, $p'_i = \varepsilon_i b_i$, and $q_i = \varepsilon_i c_i$. For a local face k , factor its induced sign $\varepsilon_i(-1)^k$ out of each of its two contributions. The remaining expressions are

k	0	1	2	3
F_k	a	a	$a + c$	c
G_k	c	$c - b$	$-b$	$-b$

The two induced face signs in a pairing are opposite by (1). Thus the column relations identify the two F values and the two G values across each paired face.

Assign formal potentials to the vertices of one tetrahedron by

$$(v_0, v_1, v_2, v_3) = (0, -b, c - b, a + c - b).$$

On each increasingly ordered face, G_k is the difference along its lower consecutive edge and F_k is the difference along its upper consecutive edge. Their sum is the difference along its remaining edge. Conversely, any additive assignment to the six oriented edges of one tetrahedron arises uniquely this way: take $a = h_{23}$, $b = -h_{01}$ and $c = h_{12}$, and use the triangle relations for the other three edges.

The presentation therefore consists of local directed-edge generators, triangle-boundary relations, and identifications of matching edges across paired faces. It is $C_1(X; \mathbb{Z})/\text{im } \partial_2$. Because X has one vertex, $\partial_1 = 0$, which proves the assertion. Repeated incidences and self-pairings merely identify some of these generators; the same presentation remains valid. $\square$

Define

$$\Gamma = \{t \in \mathbb{Z}^N : Dt \in C\mathbb{Z}^{2N} \text{ and } Dt \in \overline{C}\mathbb{Z}^{2N}\}.$$

The homomorphism

$$\alpha : \mathbb{Z}^N \longrightarrow \text{coker } C \oplus \text{coker } \overline{C}, \quad t \longmapsto ([Dt]_C, [Dt]_{\overline{C}})$$

has kernel Γ . Its cokernel is $\text{coker } K$, which vanishes by Lemma 11.1. Consequently

$$\mathbb{Z}^N / \Gamma \cong \text{coker } C \oplus \text{coker } \overline{C}. \quad (26)$$

This statement includes singular face matrices.

11.2 Saturation of the face lifts

Lemma 11.2. *The map F_0 is injective, and its image is primitive in $\mathbb{Z}^{5N}$. Equivalently, every integral vector in its rational image has an integral preimage.*

Proof. We identify its kernel over an arbitrary field k , including characteristic two. An element in $\ker(F_0 \otimes k)$ satisfies locally

$$x_1 = x_0 + x_2, \quad x_3 = x_2, \quad y_2 = y_1 + y_3, \quad y_0 = y_1, \quad y_3 = -x_0.$$

Give each ordered face the edge values $(x, x - y, -y)$ on its lower, diagonal, and upper edges. The two incident faces give the following values on the six local tetrahedron edges:

edge	first value	second value
01	x_2	x_3
02	x_1	$x_3 - y_3$
03	$x_1 - y_1$	$x_2 - y_2$
12	x_0	$-y_3$
13	$x_0 - y_0$	$-y_2$
23	$-y_0$	$-y_1$

The displayed relations make the two columns equal. Face additivity is built into the assignment, and the face variables agree across pairings. We therefore obtain a cellular cocycle on X . Conversely, recover the local face values from a cocycle by

$$(x_0, x_1, x_2, x_3) = (h_{12}, h_{02}, h_{01}, h_{01}), \quad (y_0, y_1, y_2, y_3) = (-h_{23}, -h_{23}, -h_{13}, -h_{12}).$$

The triangle cocycle relations give exactly the five local relations above. These constructions use only addition and sign changes, so they work over every field. Hence

$$\ker(F_0 \otimes k) = Z^1(X; k) = 0.$$

The last equality follows from $H_1(X; \mathbb{Z}) = 0$, the universal coefficient theorem, and the vanishing of the coboundary from the single vertex.

It follows that F_0 has column rank $4N$ over $\mathbb{Q}$ and modulo every prime. If any of its $4N$ nonzero Smith factors were divisible by a prime p , its rank modulo p would drop. All Smith factors are therefore 1, which proves both assertions. $\square$

Suppose $t, s \in \mathbb{Z}^N$ satisfy $At - Bs = 0$. Rational duality gives rational x, y such that

$$F_0 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} Dt \\ Dt \\ s - \Delta t \end{pmatrix}.$$

The right side is integral, since the diagonal entries of Δ are 0 or 1. Lemma 11.2 supplies integral x, y . Conversely, integral face data give an integral solution of $At - Bs = 0$. Thus

$$\Gamma = \Lambda, \quad \Lambda = \{t \in \mathbb{Z}^N : At \in B\mathbb{Z}^N\}. \quad (27)$$

11.3 The longitude lattice index

Lemma 11.3. *For the unhalved preferred-longitude row,*

$$\mathbb{Z}^N / (A\mathbb{Z}^N + B\mathbb{Z}^N) \cong \mathbb{Z}/2\mathbb{Z}. \quad (28)$$

Proof. We spell out the lattice normalization, since rescaling the peripheral row changes this conclusion. Let $\mathcal{T} \subset \mathbb{Z}^{3N}$ be generated by $(1, 1, 1)$ in each tetrahedron, and put $J = \mathbb{Z}^{3N} / \mathcal{T}$. The quotient map in each tetrahedron is

$$(g, g', g'') \mapsto (g - g', g'' - g').$$

Thus $J \cong \mathbb{Z}^{2N}$ is the row-coordinate lattice for $(A \mid B)$. Identify each shape coordinate with the quadrilateral type facing its opposite-edge pair; this also fixes the coordinate permutations for negative tetrahedra.

Let $Q \subset \mathbb{Z}^{3N}$ be the lattice of integral solutions of the quadrilateral matching equations, and let $\mathcal{E}$ be generated by the edge solutions. The latter are precisely the full gluing-row vectors before the quotient by $\mathcal{T}$; see [6, Sections 3.1 and 6.2]. Since $\mathcal{T} \subset Q$, the matching map descends to J , with kernel $\overline{Q} = Q / \mathcal{T}$. In particular $\overline{Q}$ is primitive in J .

Theorem 7.1 of [6] identifies

$$Q / (\mathcal{E} + \mathcal{T}) \cong \{(a, b) \in H_2(M, \partial M; \mathbb{Z}/2) \times H_1(\partial M; \mathbb{Z}) : \partial_* a = b \bmod 2\}.$$

Choose a meridian–longitude basis (μ, λ) , with μ mapping to a generator of $H_1(M; \mathbb{Z})$ and λ mapping to zero. For a knot exterior, the mod-two boundary map is injective with image generated by $\lambda \bmod 2$. Projection onto b therefore identifies this quotient with

$$L_\partial = 2\mathbb{Z}\mu \oplus \mathbb{Z}\lambda.$$

In particular, the edge lattice $\overline{\mathcal{E}} = (\mathcal{E} + \mathcal{T}) / \mathcal{T}$ is primitive in $\overline{Q}$.

The signed-corner construction for an oriented peripheral curve of class b has boundary $2b$, by [6, Section 7.1]. Our unhalved longitude row is this construction for $b = \lambda$, so its image in L_∂ is 2λ , up to a simultaneous reversal of the peripheral orientation convention. Let $L \subset J$ be generated by $\overline{\mathcal{E}}$ and this longitude row. Modulo $\overline{\mathcal{E}}$, it is $2\mathbb{Z}\lambda$, whose saturation in L_∂ is $\mathbb{Z}\lambda$. Thus

$$[\text{sat}_{\overline{Q}} L : L] = 2.$$

Because $\overline{Q}$ is primitive in J , saturation in J gives the same group. Finally, the sum of all edge rows is zero in J : this sum has the local coordinate triple $(2, 2, 2)$ in every tetrahedron. Deleting any one edge row therefore leaves the generated lattice unchanged. The rows of $(A \mid B)$ generate exactly L .

This row lattice has rank N by (4). Its index in its saturation is the product of the nonzero Smith factors of $(A \mid B)$, so that product is 2. The column cokernel of this full-row-rank matrix is consequently the group of order 2, proving (28). $\square$

11.4 The exact sequence and kernel decomposition

Define

$$g : \mathbb{Z}^N \longrightarrow \text{coker } B, \quad g(t) = [At]_B.$$

Its kernel is $\Lambda = \Gamma$, and Lemma 11.3 identifies its cokernel with $\mathbb{Z}/2\mathbb{Z}$. The map α above is surjective with the same kernel. Therefore it induces an isomorphism between $\text{coker } C \oplus \text{coker } \overline{C}$ and $\text{im } g$. Explicitly, the resulting injection is

$$\iota([Dt]_C, [Dt]_{\overline{C}}) = [At]_B.$$

Every element of its domain has this form because α is onto. If t and t' represent the same element, then $t - t' \in \Gamma = \Lambda$, so $A(t - t') \in B\mathbb{Z}^N$. This proves independence of the representative. We obtain

$$0 \longrightarrow \text{coker } C \oplus \text{coker } \overline{C} \xrightarrow{\iota} \text{coker } B \longrightarrow \mathbb{Z}/2\mathbb{Z} \longrightarrow 0,$$

as asserted in (8). No finiteness or nonsingularity assumption has been used.

For $s \in \ker_{\mathbb{Z}} B$, the integral point $(0, s)$ has an integral face lift by Lemma 11.2. Hence

$$s = Rx + \overline{R}y, \quad x \in \ker_{\mathbb{Z}} C, \quad y \in \ker_{\mathbb{Z}} \overline{C}.$$

The injectivity of F_0 makes this representation unique. This proves (7) and completes Theorem 1.3.

If C is nonsingular, nullity doubling makes B nonsingular as well. Taking finite group orders in the exact sequence and applying Theorem 10.1 gives $|\det B| = 2|\det C|^2$. If C is singular, both determinants are zero. This proves Theorem 1.4.

These arguments also apply to an oriented integral homology solid torus. Its ideal compactification has $H_1(X; \mathbb{Z}) = 0$, and its primitive homological longitude generates the kernel of $H_1(\partial M; \mathbb{Z}) \rightarrow H_1(M; \mathbb{Z})$. The same mod-two boundary calculation and the same doubled-longitude construction apply. This extension does not cover rational homology solid tori with torsion in $H_1(M; \mathbb{Z})$.

Corollary 11.4 (Odd-primary doubling and characteristic two). *At each odd prime p , the p -primary torsion of $\text{coker } B$ is the direct sum of two copies of that of $\text{coker } C$. In particular,*

$$\text{nullity}_{\mathbb{F}_p} B = 2 \text{nullity}_{\mathbb{F}_p} C \quad (p \text{ odd}).$$

In characteristic two,

$$2 \text{nullity}_{\mathbb{F}_2} C \leq \text{nullity}_{\mathbb{F}_2} B \leq 2 \text{nullity}_{\mathbb{F}_2} C + 1.$$

Proof. Localization at an odd prime kills the final term of (8); reversal identifies the first two summands. For $p = 2$, tensoring that sequence with $\mathbb{F}_2$ gives a connecting map from $\text{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/2, \mathbb{F}_2)$. Its image has dimension zero or one, which gives the two bounds. $\square$

In the broader integral-homology-solid-torus setting, the ancillary specimen **m121** has $\text{coker } C \cong \mathbb{Z}/4\mathbb{Z}$ and $\text{coker } B \cong \mathbb{Z}/4\mathbb{Z} \oplus \mathbb{Z}/8\mathbb{Z}$. It shows that (8) need not split. This specimen is not asserted to be a knot exterior in S^3 . In the singular case, the extension can also involve the free part, so the exact sequence does not by itself produce a separate 2-torsion summand.

12 An embedded relative complex and cyclic face torsion

We prove Theorem 1.6 by realizing $C^{\top}$ as a relative cellular boundary map. Throughout this section M may be any oriented integral homology solid torus with a one-cusped ordered ideal triangulation. No angle structure is required.

12.1 The local disks

Truncate the ideal vertices. In each quotient face f , choose an arc u_f parallel to its lower edge, oriented from its lowest to its middle cusp corner. Its two endpoints lie in the interiors of the corresponding truncation edges. Make the choices on quotient faces and lift them to local faces; this ensures agreement across every face pairing.

In one truncated tetrahedron, form two closed curves on its boundary. The first follows $u_2, u_0, -u_1$, joined in the cusp caps at vertices 1, 2, 0. The second follows $u_2, -u_3$, joined in caps 1, 0. More explicitly, the required joining arcs are

cap	first curve	second curve
0	sides 1 and 2	sides 2 and 3
1	sides 2 and 0	sides 2 and 3
2	sides 0 and 1	none
3	none	none

Here a side is labeled by the face containing it. In caps 0 and 1, the two arcs can be chosen with disjoint interiors and with their common endpoint on side 2. All other joining arcs lie in different cap interiors. The two closed curves consequently share exactly u_2 and form an embedded theta graph on the boundary sphere.

The two cycles containing u_2 bound the two complementary disk regions incident to that edge. Choose a collar of the boundary sphere inside the tetrahedron. On each of these disk regions choose a piecewise linear height function that is zero on the boundary and positive in the interior. The graphs of the two functions in the collar are embedded disks with disjoint interiors, meeting exactly along u_2 . This construction fixes the boundary curves and moves all disk interiors into the tetrahedron interior. Orient the disks independently so that, modulo their cusp arcs, their boundaries are

$$u_2 + u_0 - u_1, \quad u_2 - u_3.$$

Glue these patches across the paired faces and denote their union by $\mathcal{B}_C \subset M$. The result is an embedded finite PL complex. Indeed, disk interiors lie in tetrahedron interiors, which are not identified by face gluings. At a quotient face, pages from the two tetrahedral sides lie in its two distinct half-neighborhoods and meet only along u_f . At an endpoint of u_f , exactly two cusp-triangle sides are paired. Since the endpoint is in the interior of such a side, there are no additional identifications through cusp vertices. These descriptions also apply when two distinct faces of the same tetrahedron are paired. Individual cell closures may acquire boundary identifications; the resulting cellular structure need not be regular. Set

$$\mathcal{G}_C = \mathcal{B}_C \cap \partial M.$$

This is the embedded graph formed by the cusp arcs.

12.2 The relative cellular complex

The relative cells of $(\mathcal{B}_C, \mathcal{G}_C)$ are the $2N$ face arcs u_f in dimension one and the two disks per tetrahedron in dimension two. There are no relative zero-cells and no cells of higher dimension. With the first disk of each tetrahedron listed before the second family, the boundary map is $C^\top$. Therefore

$$H_1(\mathcal{B}_C, \mathcal{G}_C; \mathbb{Z}) = \text{coker } C^\top, \quad H_2(\mathcal{B}_C, \mathcal{G}_C; \mathbb{Z}) = \ker C^\top. \quad (29)$$

The complex $\mathcal{B}_C$ is connected: each tetrahedral patch is connected and includes all four local face arcs, and adjacent tetrahedral patches share the arc of their paired face.

12.3 Filling and divisibility

Choose a solid-torus filling $Y = M \cup V$ that is an integral homology sphere. For a knot exterior in S^3 , take its original meridional filling. Such a filling also exists for every integral homology solid torus: Poincaré–Lefschetz duality gives $H_1(M, \partial M; \mathbb{Z}) = H^2(M; \mathbb{Z}) = 0$, so the boundary map onto $H_1(M; \mathbb{Z}) = \mathbb{Z}$ is surjective. Any slope mapping to a generator is primitive, and filling along it gives $H_1(Y; \mathbb{Z}) = 0$ by the Mayer–Vietoris sequence. Closed oriented three-manifold duality then makes Y an integral homology sphere.

Put

$$\mathcal{Z} = \mathcal{B}_C \cup V \subset Y.$$

Since $\mathcal{B}_C \cap V = \mathcal{G}_C$, the relative cellular complex of $(\mathcal{Z}, V)$ is that of $(\mathcal{B}_C, \mathcal{G}_C)$.

We need the following elementary consequence of duality: the first integral homology of an embedded finite PL complex in an integral homology three-sphere is free abelian. In the present case, choose a closed regular neighborhood U of $\mathcal{Z}$ in Y and put $W = \overline{Y \setminus U}$. The regular neighborhood retracts onto $\mathcal{Z}$. The pair sequence for (Y, U) , excision across collars, and Poincaré–Lefschetz duality give

$$H_1(\mathcal{Z}; \mathbb{Z}) \cong H_1(U; \mathbb{Z}) \cong H_2(Y, U; \mathbb{Z}) \cong H_2(W, \partial W; \mathbb{Z}) \cong H^1(W; \mathbb{Z}).$$

The pair-sequence isomorphism uses $H_2(Y) = H_1(Y) = 0$. Finally, $H^1(W; \mathbb{Z})$ is free abelian by the universal coefficient theorem; this does not require W to be connected.

Both V and $\mathcal{Z}$ are connected. Their pair sequence and (29) consequently give

$$\text{coker } C^T \cong H_1(\mathcal{Z}; \mathbb{Z}) / \text{im } H_1(V; \mathbb{Z}). \quad (30)$$

The image of $H_1(V; \mathbb{Z})$ is generated by the class of the filling core. Write $H_1(\mathcal{Z}; \mathbb{Z}) \cong \mathbb{Z}^r$ and let v be this core class. If $v = 0$, the quotient is free. Otherwise, write $v = dv_0$ with v_0 primitive and $d > 0$. Extending v_0 to an integral basis gives

$$\mathbb{Z}^r / \mathbb{Z}v \cong \mathbb{Z}^{r-1} \oplus \mathbb{Z}/d\mathbb{Z}.$$

Thus the torsion subgroup of $\text{coker } C^T$ is cyclic. Transposition preserves Smith normal form, so the same is true of $\text{coker } C$.

If C is nonsingular, (29) also gives $H_2(\mathcal{Z}, V; \mathbb{Z}) = 0$. The pair sequence then makes $H_1(V; \mathbb{Z}) \rightarrow H_1(\mathcal{Z}; \mathbb{Z})$ injective. Its cokernel is finite, so $H_1(\mathcal{Z}; \mathbb{Z})$ has rank one and the core class is nonzero. Its divisibility d satisfies

$$\text{coker } C^T \cong \mathbb{Z}/d\mathbb{Z}, \quad d = |\det C|.$$

This proves Theorem 1.6.

For nonsingular C , unimodularity is therefore equivalent to primitivity of this filling-core class. The argument does not prove that primitivity. In particular, rank-one homology does not imply that a regular neighborhood of $\mathcal{Z}$ is a solid torus. A further restriction from the geometry or ordering would be needed to deduce the original unit-determinant assertion.

13 Computational checks and remaining questions

13.1 Reproducibility and scope of the checks

The ancillary files contain executable exact-arithmetic checks, fixed input data, recorded output and pinned package versions. The checks reconstruct face matrices from ordered face tables, independently of the FAMED implementation used in [1]. SnapPy supplies the gluing exponents and the

fundamental-group presentation. Abelianization identifies the primitive rational longitude; for the knot exteriors this is the preferred longitude. The checks were run with Python 3.13, SnapPy 3.3.2, Regina 7.4.1 and SymPy 1.14.0.

Check	Extent
Local cochains	48 polynomial identities, including both tetrahedron signs
Affine coordinates	The figure-eight example, including its logarithmic constants
Fixed specimens	10 ordered triangulations, including face nullities 0, 1, 2
Small face pairings	All 10,503 labelled ordered tables with at most three tetrahedra

The fixed-specimen checks include the entire projected face relation, the constrained symmetric forms, integer-kernel decompositions, Smith factors, the determinant formula and reductions modulo 2, 3, 5, 7, 11. Some specimens are integral homology solid tori with nonunit face determinant. They are not asserted to be knot exteriors in S^3 and therefore do not contradict the determinant conjecture in [1].

Of the 10,503 labelled tables, 712 give connected valid orientable one-torus ideal triangulations with first Betti number one. Among these, 43 have no edge of degree one or two, and 31 admit a strict angle structure. The positive-rank check passes on every table in these latter classes. These are counts of labelled tables, not counts of distinct manifolds. Regina’s strict-angle test checks an angle structure; it is not a certificate of a complete geometric shape solution.

The ancillary `README.md` gives four commands that reproduce these checks. The two regenerated main result files agree byte-for-byte with the recorded outputs. Ranks, kernels, determinants, and matrix identities are decided by exact arithmetic. Numerical shape approximations play no role in these decisions. The finite computations provide checks on conventions and possible counterexamples; the universal claims depend on the proofs above.

13.2 Questions left open

Theorem 1.4 reduces the remaining determinant assertion in Conjecture 4.1 of [1] to $|\det C| = 1$ in its nonsingular geometric knot setting. Section 12 interprets this condition as primitivity of a filling-core class, but does not establish that primitivity. Proving it requires information beyond the cyclicity of the face torsion.

Wong’s longitude–meridian conditions in Definition 1.5 of [3] remain additional hypotheses for the Jones-function results. The preferred-longitude relation proved here does not supply these additional conditions. Finally, this work does not construct a suitable geometric triangulation for every hyperbolic knot. The full Andersen–Kashaev volume conjecture consequently remains outside the conclusions of this paper.

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A AI tools and verification status

OpenAI Codex was used extensively to propose and develop the mathematical arguments, draft and revise the manuscript, generate computational checks, search the literature, and conduct internal reviews. The author directed the investigation and preparation of the manuscript. The use of Codex extended to the mathematical development and was not limited to language editing.

The arguments have not been independently verified by a human specialist or a formal proof assistant. Internal reviews and finite exact-arithmetic checks do not establish correctness or novelty. This preprint is circulated for critical scrutiny; responsibility for its contents rests with the named author.